Editorial Office
Sensors

Dear Editors,

Please consider our manuscript, "*Robust Stackelberg-Supervised Overtaking Policy with Persistent Tactical Corridors for Autonomous Racing*," for publication as an Article in Sensors.

High-speed overtaking requires tactical decisions that remain consistent as vehicle interactions evolve and can be executed within geometric and dynamic constraints. Our study addresses this challenge by representing a learned overtaking decision as a persistent tactical feasible region. An offline robust Stackelberg response-set teacher supervises a reinforcement learning policy, while a hysteretic finite-state machine and deterministic corridor construction preserve passing-side commitment and adapt the admissible region to the evolving track and opponent geometry. This region connects tactical reasoning to constrained path planning, speed profiling, and nonlinear model predictive control, retaining flexibility for repeated replanning.

We evaluate the complete stack in hardware-in-the-loop experiments using the EAV24 simulator and the target onboard computer, with noise and packet drops applied to localization and perception outputs. Across 1,000 laps in the primary evaluation, the method achieved a 94.7% lead-acquisition rate and a mean time to lead acquisition of 4.82 s, with no collisions or track-boundary violations observed. Its lead-acquisition rate exceeded those of three baselines sharing the same execution stack by 31.2–66.2 percentage points. With the policy and execution parameters unchanged, lead-acquisition rates were 91.8% on Imola and 88.9% on Suzuka, two previously unseen circuits.

We believe this work will interest readers of Sensors working on sensor-based autonomous systems, intelligent vehicles, and decision-making under imperfect state estimates. It provides a concrete interface between learned tactical decisions and constrained vehicle execution, supported by evaluation on the target computing hardware.

Thank you for considering our manuscript.

Sincerely,

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